

PoE setups and components usually have a 10 to 20 per cent higher price tag, compared to traditional hardware. However, installation cost and overall operational efficiency make up for this higher expense.

Thukral continues, "Collaterally, a single power and Ethernet source means easier maintenance. Only one backup unit is sufficient for all services over a PoE cable. There is no need for installing multiple local power outlets either."

All these lead to an eventual cost optimisation of 30 to 50 per cent in setup, operation and maintenance.

Higher reliability. Normal power cables are prone to wear-and-tear and damage. PoE cables are comparatively resilient. There is a strong safety factor with PoE. A low-voltage test current is initially passed through PoE cables to test resistance and safety standards in place. Full power is channelled only after ensuring this.

On the other hand, wireless communication is prone to signal interference. This can be problematic in case of essential devices such as surveillance cameras, data loggers and routers. PoE cables keep the risk of interference at bay.

There is a certain reduction of downtime as well. Thukral says, "Users get a chance to keep an eye on the operation of all connected devices in real time. It is easier for them to manage the common point in PoE compared to distributed devices, making troubleshooting easier." PoE systems can usually be invigilated and controlled through graphical user platforms.

Flexibility. Users can retrofit the installation on their existing non-PoE appliances as well. For instance, if they own a general (non-PoE) network switch but have a PoE-enabled powered device, then a PoE injector acts as a bridge between the two, adding the power line along with data stream from the switch.

On the other hand, if a single powered device in the setup is not PoE-compliant, while a source PoE switch is available, a PoE splitter can separate the power line from data coming from the switch and run the powered device as required. Moreover, PoE is a scal-

able solution and allows powering of appliances across long distances.

Final word

PoE has been picking up popularity for a few years now. While buying the hardware can be comparatively costly initially, application in the right areas of business can derive a quick and assured ROI.

—Paromik Chakraborty, technical journalist, EFY

Robots: Giving An Edge To Smart Workplaces

➤ **The biggest benefit robots bring to organisations is cost optimisation**

The robotic army is taking over our workplaces. The presence of robots is no more limited to factories and warehouses to lift heavy loads. It has spread to more sophisticated areas for assisting human beings, beyond their own analytical

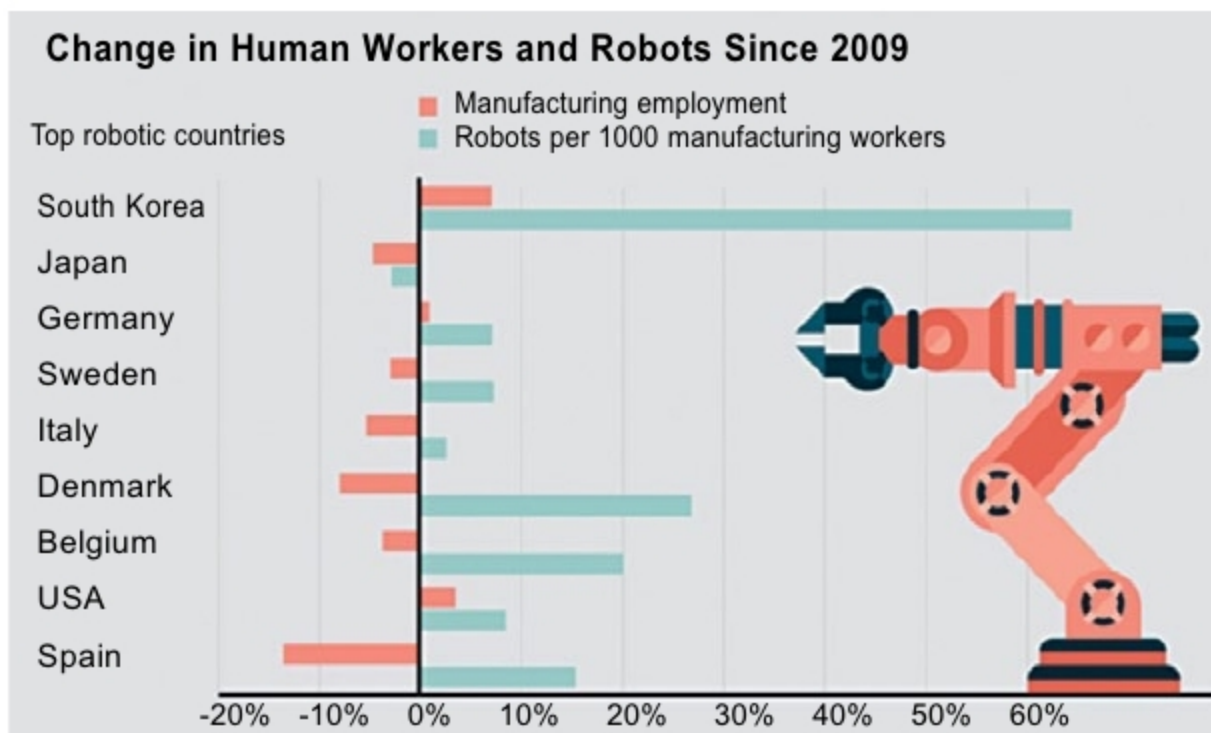
abilities, and for making decisions based on facts.

Sumit Joshi, vice chairman and managing director, Philips Lighting India, shares, "India will have the youngest employable population in the world by 2020. With an average age of 29, it is predicted that, as flexibility and ease of work goes up, the average retirement age will also go up. This will result in a workforce that is multi-generational and more diverse than ever before. Thus, workplaces for a resurgent and young India will need to be designed to keep employee comfort, well-being, energy-efficiency and connectivity in mind." Therefore as India strives to create advanced workspaces, robots are being integrated in to the core of systems.

Amarendra Srivastava, a consultant for smart workplaces, shares, "Robots have been present on industrial floors for decades now. But, earlier they were considered helpers. In modern workplaces, robots are taking up more diverse and collaborative roles with humans. They make the workspaces more interactive and increase mobility."

Roles robots play

Chetan Chawla, digital transformation consultant, Emaar, says, "Currently, robots are mostly in demand for jobs that are repetitive in nature. These jobs could be mechanical or technical. It is believed that artificial intelligence (AI)-embedded robots will be able to



Robots taking up human roles and changing work environment (Credit: www.roboticstomorrow.com)